Sao Aphisith Sithisack

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EDUCATION

University of Oklahoma — **Norman, OK**
B.S. in Electrical & Computer Engineering (Honors) | Minor in Mathematics & Computer Science | GPA: 4.0 — *Aug 2024 – May 2028*

RELEVANT EXPERIENCE

Undergraduate Research Assistant — **Jan 2026 – Present**
Pulsed Laser Template Engineering (PLATEN) — *Norman, OK*
- Fabricate **high-aspect-ratio (HAR) micro- and nanostructures** via photolithography, reactive ion etching (RIE), and pulsed laser deposition (PLD).
- Optimize anisotropic plasma etch parameters (gas chemistry, RF power, pressure) to achieve vertical sidewalls and controlled feature depth.
- Support end-to-end process development including resist patterning, etch selectivity, and post-process inspection.

Electrical Systems Member — **Jan 2026 – Present**
Sooner Competitive Robotics — *Norman, OK*
- Designed power distribution architecture and CAD layouts for electrical component placement and wire routing.

Peer Learning Assistant — **Aug 2025 – Present**
Student Learning Center — *Norman, OK*
- Mentored 250+ students per semester in Physics I–II, Calculus I–III, and General Chemistry I using structured quantitative problem-solving methods.

Research Apprentice — **Aug 2025 – Dec 2025**
Light–Matter Interaction Research (Dr. Tischler) — *Norman, OK*
- Reviewed light–matter interactions in silicon carbide nanocrystals and gained exposure to optical spectroscopy systems.

Research Apprentice — **Jan 2025 – May 2025**
First Year Research Experience – Cloud Microphysics Group — *Norman, OK*
- Modeled atmospheric profiles using Python (NumPy, Pandas, Matplotlib) with numerical methods and statistical analysis.

PROJECTS

Electrical Engineering Planner — **Apr 2026**
Personal Project — *GitHub*
- Designed a structured project planner that curates domain-specific engineering curricula across VLSI, Hardware Engineering, Microelectronics, and Photonics.
- Integrated Python-based CMOS simulations (NumPy, Matplotlib) and industry EDA tools (Cadence Virtuoso, Magic VLSI, Vivado) to guide hands-on IC design projects from concept to implementation.

Job Application Agent — **Mar 2026 – Present**
Personal Project — *GitHub*
- Built an autonomous 7-phase Python pipeline that parses resumes, scores jobs across 6 weighted criteria, tailors application materials, and generates Excel trackers with 3 pluggable LLM backends (Claude API, Ollama, offline regex).

Climb – Hacklahoma Project — **Feb 2026**
Personal Project — *GitHub*
- Pomodoro productivity app built with Expo, React Native, and TypeScript featuring a reward/penalty system for app usage restriction.

ADDITIONAL EXPERIENCE

Alumni Coordinator / Board Member — **Jul 2024 – Present**
United World College Lao PDR National Committee — *Hybrid*
- Led fundraising exceeding $2,500 and coordinated communication among alumni, scholars, and partner organizations.

Sponsorship & Volunteering Chair — **Nov 2024 – Present**
Association of Southeast Asian Nations at OU — *Norman, OK*
- Secured $1,000+ in external funding through outreach to 20+ donors and organized cultural events.

TECHNICAL SKILLS

Python (NumPy, Pandas, Matplotlib) | Statistical Analysis | Data Visualization | AI-Assisted Data Analysis | LLM Integration (Claude API, Ollama) | Optical Spectroscopy | Spectrometers | Pulsed Laser Deposition (PLD) | Numerical Methods | MATLAB | LabVIEW | Photolithography | Anisotropic RIE | Cleanroom Processing | CAD Layout Design | LTSPICE | Java

INTERESTS

Photonics | Optoelectronics | Semiconductor Manufacturing | High-Aspect-Ratio Devices | Microelectronics | VLSI Design | Integrated Circuit Design | Hardware Engineering